



PRODUCT NAME

High-performance Hair Cream Formula

Description

A professional hair cream formulation designed to deliver moisture while maintaining a non-greasy finish. Easy to apply, it provides high performance for hair care enthusiasts and professionals alike.

Technical Specifications

pH Level:	7
Viscosity:	High
Shelf Life:	12 months
Batch Size:	10 kg - 100 kg
Processing Time:	2 hours
Temperature:	75°C for emulsifying; cool to 40°C before adding sensitive ingredients
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Homogenizer, heating vessels, stirrers
Certification:	ISO 9001:2015

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	60.0%	Solvent
2	Cetyl Alcohol	Cetyl Alcohol	5.0%	Emollient, Thickener
3	Stearic Acid	Stearic Acid	4.0%	Emulsifying Agent
4	Glycerin	Glycerin	10.0%	Humectant
5	Ceteareth-20	Ceteareth-20	3.0%	Emulsifier
6	Sunflower Seed Oil	Helianthus Annuus (Sunflower) Seed Oil	5.0%	Moisturizing Agent
7	Dimethicone	Dimethicone	3.0%	Conditioning Agent
8	Fragrance	Parfum	1.0%	Fragrance
9	Phenoxyethanol	Phenoxyethanol	1.0%	Preservative
10	Sodium Hydroxide	Sodium Hydroxide	0.1%	pH Adjuster
11	Xanthan Gum	Xanthan Gum	0.9%	Thickening Agent
12	Hydrolyzed Silk	Hydrolyzed Silk	1.0%	Conditioning Agent
Total			100%	

Manufacturing Process

Phase 1: Phase A

- Heat Water to 75°C.
- Add Glycerin and Xanthan Gum to Water while stirring to ensure full dispersion.

Phase 2: Phase B

1. Combine Cetyl Alcohol, Stearic Acid, and Cetearth-20 in a separate vessel.
2. Heat to 75°C until all components are fully melted.
3. Add Dimethicone and Sunflower Seed Oil to the melted mixture.

Phase 3: Phase C

1. Once both Phase A and B are at 75°C, slowly add Phase B to Phase A under continuous stirring using a homogenizer.
2. Cool down the mixture to 40°C while maintaining stirring.

Phase 4: Phase D

1. Add Phenoxyethanol, Fragrance, and Hydrolyzed Silk as the mixture cools.
2. Adjust pH to 7 using Sodium Hydroxide solution if required.

Required Equipment

Homogenizer, heating vessels, stirrers

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: ISO 9001:2015

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 10 kg - 100 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a small amount of cream to damp or dry hair and style as desired. Focus on the ends and dry areas for best results.